

Customer Shopping Behavior Analysis & Data Pipeline Project Report

1.1 Project Overview

This project will help the leading retail company to understand their behavior on their shopping habits on a regular basis so that they can act according to the desires of the consumer and analyzing which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases. The main motive of this analysis is to leverage customers' shopping data and search for the trends and patterns to improve customer engagement.

1.2 Executive Summary

This report documents the end-to-end data cleaning, feature engineering, and database integration process performed on the **Customer Shopping Behavior** dataset (3,900 records). The goal was to transform raw transactional records into a structured, clean, and database-ready format to enable downstream analytics, customer segmentation, and strategic marketing insights.

1.3 Data Ingestion & Initial Exploratory Analysis

The raw dataset was loaded into a Pandas DataFrame from customer_shopping_behavior.csv containing 3,900 customer records and 18 initial variables.

- **Initial Dataset Dimensions:** 3,900 rows × 18 columns.
 - **Key Attributes Identified:** Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Color, Season, Review Rating, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Previous Purchases, Payment Method, Frequency of Purchases.
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1.4 Data Cleaning & Transformation Pipeline

Step 1: Missing Value Assessment & Category-Based Imputation

While treating the dataset firstly the missing values were analyzed and treated. For which `.isnull().sum()` was used to see total numbers of missing data in each columns.

```
#to find the missing values  
df.isnull().sum()
```

```
Customer ID          0  
Age                  0  
Gender                0  
Item Purchased        0  
Category              0  
Purchase Amount (USD) 0  
Location              0  
Size                  0  
Color                 0  
Season                0  
Review Rating        37  
Subscription Status   0  
Shipping Type         0  
Discount Applied      0  
Promo Code Used       0  
Previous Purchases    0  
Payment Method        0  
Frequency of Purchases 0  
dtype: int64
```

Through this result, 37 review rating were missing in the dataset. For the dataset to be evaluated correctly we need to replace the missing value with the average review rating of each category and replace the nan values with the median values.

```
# replacing the nan to median review rating for each category
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x:x.fillna(x.median()))
```

```
df.isnull().sum()
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

Step 2: Column Standardization & Normalization

To adhere to database naming standards (SQL/PostgreSQL conventions), column headers were normalized by converting strings to lowercase, replacing spaces with underscores, and simplifying complex names:

- Converted all header text to lowercase (e.g., Customer ID → customer_id).
- Replaced spaces with underscores (_).
- Renamed purchase_amount_(usd) to purchase_amount.

```
df.columns=df.columns.str.lower()
df.columns= df.columns.str.replace(' ', '_')
df=df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

Step 3: Demographic Binning & Feature Engineering

To prepare data for targeted demographic marketing strategies, the continuous age attribute (ranging from 18 to 70 years, median: 44) was partitioned into four quantile-based age groups using `pd.qcut()`:

```
#categorize the customer according to their age group
lables= ['Young Adults', 'Adults', 'Middle Aged', 'Senoir']
df['age_group']= pd.qcut(df['age'], q=4, labels= lables)
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle Aged
1	19	Young Adults
2	50	Middle Aged
3	21	Young Adults
4	45	Middle Aged
5	46	Middle Aged
6	63	Senoir
7	27	Young Adults
8	26	Young Adults
9	57	Middle Aged

Step 4: Categorical Mapping & Numeric Conversion

The `frequency_of_purchases` feature contained textual purchase cadences. These string descriptors were mapped to numerical values representing cycle duration in days (`purchase_frequency_days`) to enable numerical modeling:

```
#purchase freq days
df['frequency_of_purchases'].unique()
```

```
<StringArray>
[   'Fortnightly',      'Weekly',      'Annually',      'Quarterly',
    'Bi-Weekly',      'Monthly', 'Every 3 Months']
Length: 7, dtype: str
```

```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Annually': 365,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Monthly': 30,
    'Every 3 Months': 9
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
df[['purchase_frequency_days', 'frequency_of_purchases']].head()
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually

Step 5: Redundancy Analysis & Feature Elimination

A comparison between `discount_applied` and `promo_code_used` was conducted. To streamline the schema and reduce feature redundancy, `promo_code_used` was dropped from the dataset.

```
df['discount_applied'] == df['promo_code_used'].all()
```

```
0      False
1      False
2      False
3      False
4      False
...
3895   False
3896   False
3897   False
3898   False
3899   False
Name: discount_applied, Length: 3900, dtype: bool
```

```
df= df.drop('promo_code_used', axis=1)
```

Step 6. Database Integration & Persistence

The final cleaned dataset was loaded directly into a relational database via SQLAlchemy and PostgreSQL.

- **Database Engine:** PostgreSQL (via psycopg2 driver & SQLAlchemy engine).
- **Target Table:** customer (Database: customer_behaviour).
- **Execution Method:** df.to_sql('customer', engine, if_exists='replace', index=False).
- **Final Table Schema:** 19 columns including engineered features (age_group and purchase_frequency_days).

```
from sqlalchemy import create_engine
#placeholders to connect the database
username = "postgres"
password = "2063"
host = "localhost"
port = "5432"
database = "customer_behaviour"

#creating an engine
engine= create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

table = "customer"
df.to_sql(table, engine, if_exists="replace", index=False)
print("sucessfully Connected!!")
```

```
sucessfully Connected!!
```

1.5 Data Analysis using SQL

The structured analysis performed in PostgreSQL to answer the business questions:

1. **Revenue by gender:** The total revenue generated through the male and the female customers is compared.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users:** Identified the customers who used the discounts but still spends above the average purchase amount

	customer_id bigint	purchase_amount bigint	average numeric
1	475	76	76
2	878	65	65
3	1517	68	68
4	1360	68	68
5	654	68	68
6	627	79	79
7	128	89	89
8	983	62	62
9	1198	92	92
10	1365	75	75
11	461	65	65
12	1659	66	66
13	389	82	82
14	902	68	68
15	1390	72	72
16	1019	70	70
17	862	100	100
18	1466	72	72
19	484	73	73
Total rows: 839		Query complete 00:00:00.168	

3. **Highly Rated products:** Searched and found the products that are highly rated.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison:** Compared the average the purchase amounts between Standard and Express shipping.

	shipping_type text	amount numeric
1	Express	60.48
2	Standard	58.46

5. **Subscriber vs Non-Subscribers:** Compared the average spend and total revenue across subscription status to see if the subscribed customer spend more.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products:** Searched for top 5 products have the highest percentage of purchases with discount.

	item_purchased text	discount rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

- 7. Categorizing the customers:** Segmenting the customers into new, returning and loyal group according to the total number of previous purchases.

	Customer Segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

- 8. Top 3 most purchased products in each category:** Finding the most purchased products in each category to find the trend in the products.

	Rank bigint	Category text	Item Purchased text	Total Orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

- 9. Subscription trend:** Finding the subscription trend in the repeat buyers.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue from each age group: Evaluating the revenue impact from each age group.

	age_group text	sum numeric
1	Young Adults	62143
2	Middle Aged	59197
3	Adults	55978
4	Senoir	55763

1.6 Conclusion

- Customer behavior varies across age, gender, spending, and purchase frequency.
 - Discounts play an important role in influencing purchasing behavior, including among high-spending customers.
 - Returning and loyal customers are valuable segments for customer retention and targeted marketing.
 - Analysis of top-selling and highly rated products helps identify customer preferences and product demand.
 - Purchase frequency can be used to develop timely and personalized repurchase reminders.
 - Repeat buyers provide a potential opportunity for subscription-focused marketing strategies.
 - Overall, the findings support a data-driven and targeted approach to customer engagement and retention.
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1.7 Business Recommendations

- Implement targeted marketing campaigns based on customer age groups and purchasing behavior.
 - Use personalized promotions for new, returning, and loyal customers.
 - Introduce automated purchase reminders based on customers' purchase frequency.
 - Optimize discount strategies by identifying products and customer segments that respond strongly to discounts.
 - Promote highly rated and popular products to increase customer engagement and sales.
 - Develop subscription and loyalty offers specifically for repeat customers.
 - Continuously analyze customer behavior to support data-driven marketing and retention strategies.
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